



19EC412 - COMMUNICATION NETWORK

SKILL ASSESSMENT 1

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DEPT : ECE
YEAR : 3 RD YEAR

MESH TOPOLOGY USING CISCO PACKET TRACER

1. Introduction

A Mesh Topology is a network structure in which every networking device is connected directly to every other device. It provides multiple communication paths, ensuring high reliability and uninterrupted communication.

Mesh topology is commonly used in communication systems because it provides:

- High Reliability
- Fault Tolerance
- Better Security
- Multiple Communication Paths
- Fast Data Transmission

It is widely used in:

- Banking systems
- Military communication
- Telecommunication networks
- Data centers
- Enterprise networks

2. Concept of Mesh Topology

In mesh topology:

- Every switch or node is directly connected to all other switches.
- Multiple communication paths are available.
- Failure of one connection does not affect the entire network.
- Data can travel through alternate routes.

There are two types of mesh topology:

- Full Mesh Topology
- Partial Mesh Topology

In this project, a Full Mesh Topology is implemented using switches and PCs.

- Switch0 → PC0
- Switch1 → PC2
- Switch2 → PC1
- Switch3 → PC3
- Switch4 → PC4
- Switch5 → PC5

Total Devices

- 6 PCs
- 6 Switches

4. Diagram Representation

Explanation

Mesh Structure

- All switches are interconnected using dedicated links.
- Multiple communication paths exist between devices.

Device Communication

- Each PC communicates through its connected switch.
- Switches forward data using direct mesh links.

5. Components Required

Component	Quantity	Purpose
PCs	6	End Devices
Switches	6	Network Communication
Copper Straight Through Cable	6	PC-to-Switch Connection
Copper Cross-Over Cable	Multiple	Switch-to-Switch Connection
Cisco Packet Tracer	1	Network Simulation

6. Working Principle

1. Each PC sends data to its connected switch.
2. The switch checks the destination address.
3. Data is forwarded directly through mesh links.
4. Multiple paths are available for transmission.
5. If one link fails, another path is used automatically.

6. The destination switch forwards data to the target PC.

7. Example Scenario

Suppose:

PC0 wants to send data to PC5.

Data Flow:

PC0 → Switch0 → Switch5 → PC5

Alternative paths may also be used if one connection fails.

This demonstrates reliable communication using mesh topology.

8. Real-Time Scenario

Banking Network Example

- Bank branches are connected using mesh topology.
- Secure communication is maintained between branches.
- Backup paths improve reliability.

Benefits

- Secure communication
- Continuous network availability
- High reliability
- Better fault tolerance

9 Advantages

- High Reliability
- Better Fault Tolerance
- Multiple Communication Paths
- Fast Data Transmission
- Better Security
- Failure Isolation

10. Disadvantages

- High Installation Cost
- Complex Network Design
- Large Number of Cables Required
- Difficult Maintenance

- Poor Scalability for Large Networks

11. Applications

Mesh topology is used in:

- Banking systems
- Military communication
- Telecommunication industries
- Internet backbone networks
- Wireless mesh networks
- Enterprise networks

12. Comparison with Other Topologies

Feature	Star Topology	Bus Topology	Mesh Topology
Reliability	Medium	Low	High
Cost	Medium	Low	Very High
Scalability	Moderate	Low	Moderate
Complexity	Low	Low	High
Performance	Good	Average	Excellent

13. Conclusion

Mesh topology is one of the most reliable networking structures used in modern communication systems. In this project, a Full Mesh Topology was implemented using six switches and six PCs interconnected through dedicated communication links.

The topology successfully provides:

- Direct communication paths
- High fault tolerance
- Secure data transmission

This design improves:

- Reliability
- Network performance
- Communication security

Although mesh topology requires more cables and higher installation cost, its reliability and fault tolerance make it highly suitable for critical communication systems such as banking, military, and enterprise networks.

Thus, the implementation of Mesh Topology using Cisco Packet Tracer provides an efficient and reliable networking solution.

14. Important Questions

Fill in the Blanks

1. Mesh topology provides multiple _____ paths.
2. Every node in full mesh is connected to _____ devices.
3. Mesh topology provides high _____ tolerance.
4. Dedicated links improve network _____.
5. Mesh topology mainly uses _____ connections.

15. Match the Following

Column A	Column B
Mesh Topology	Dedicated Links
Reliability	Fault Tolerance
Direct Communication	Point-to-Point
Switch	Data Forwarding

Answer Key

- 1 – Dedicated Links
- 2 – Fault Tolerance
- 3 – Point-to-Point
- 4 – Data Forwarding

16. True or False

1. Mesh topology uses multiple communication paths. — True
2. Failure of one link affects the entire network. — False
3. Mesh topology provides high reliability. — True
4. Switches are used for communication in this project. — True
5. Mesh topology requires fewer cables. — False

17. Multiple Choice Questions (MCQs)

1. Mesh topology mainly provides:

- a) Low security
- b) High reliability
- c) Single connection
- d) No communication

Answer:b

2. In full mesh topology:

- a) Devices are partially connected
- b) No cables are used
- c) Every device connects to every other device
- d) Only one switch is used

Answer:c

3. The major disadvantage of mesh topology is:

- a) Low reliability
- b) Poor communication
- c) High installation cost
- d) No security

Answer:c

4. Mesh topology mainly uses:

- a) Backbone cable
- b) Dedicated links
- c) Wireless tower
- d) Single bus line

Answer:b

5. Mesh topology is suitable for:

- a) Critical communication systems
- b) Offline systems
- c) Temporary networks
- d) Small isolated computers

Answer:a

18. Short Answer Questions

1. What is mesh topology?
2. Explain full mesh topology.
3. What are the components required?
4. Explain the working principle of mesh topology.
5. Mention two advantages of mesh topology.

19. Long Answer Questions

1. Explain mesh topology with a neat diagram.
2. Discuss the working principle of mesh topology.
3. Compare mesh topology with star and bus topology.
4. Explain the advantages and disadvantages of mesh topology.
5. Describe a real-time application of mesh topology.